

Sayan Bhowmik

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RESEARCH INTERESTS

Developing real-space density functional theory (DFT) methods for electrochemical and catalytic applications, with emphasis on surface and interface modeling via bulk boundary condition formalisms. Contributions span core algorithm development in the SPARC electronic structure code, including DFT+U and spectral scheme for atomic structure calculations. Previous experience span studying defect physics in aluminum and nanopore engineering in 2D materials.

EDUCATION

Georgia Institute of Technology <i>Ph.D. in Chemical & Biomolecular Engineering (GPA: 3.88/4.0)</i>	Atlanta, USA <i>Expected 2027</i>
Indian Institute of Science (IISc) <i>M.Tech (Gold medal) in Chemical Engineering (CGPA: 9.62/10)</i>	Bengaluru, India <i>2022</i>
Jadavpur University <i>B.E. (Class Rank 3) in Chemical Engineering (CGPA: 8.99/10)</i>	Kolkata, India <i>2020</i>

RESEARCH EXPERIENCE

Georgia Institute of Technology <i>Graduate Research Assistant · Advisors: Dr. Phanish Suryanarayana & Dr. Andrew J. Medford</i>	Atlanta, USA <i>2022 – Present</i>
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- **Bulk Boundary Conditions for Electrochemical Simulations (Current):** Developing a real-space DFT formalism exploiting nearsightedness of electronic interactions to restrict calculations to a localized surface region, enabling accurate surface and adsorption energy modeling for catalytic applications.
- **Real-Space DFT+U with Spectral Atomic Solver:** Developed a spectral solver achieving $\sim 1 \mu\text{Ha}$ eigenvalue accuracy to generate XC-consistent atomic projectors, enabling a large-scale parallel DFT+U framework that outperforms plane-wave codes by over an order of magnitude in time to solution for large systems; introduced a hybrid-functional scheme for Hubbard parameter optimization.
- **Strain-Driven Defect Engineering in Aluminum:** Identified compressive strain as a key driver of vacancy clustering and prismatic dislocation loop nucleation on the (111) plane via DFT calculations.

Indian Institute of Science (IISc) <i>M.Tech Research Fellow · Advisor: Dr. Ananth Govind Rajan</i>	Bengaluru, India <i>2021 – 2022</i>
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- **Nanopore Formation in 2D MoS₂:** Combined DFT etching energies, Marcus theory, kinetic Monte Carlo, and chemical graph theory to elucidate nucleation mechanisms and catalog nanopore isomers, establishing chemical environment as a tunable handle for controlling nanopore geometry and size distribution.

PUBLICATIONS

Lead author: 4 (PhD), 3 (M.Tech), 1 (B.E.); Contributing author: 1 (PhD)

1. **S. Bhowmik**, A. J. Medford, P. Suryanarayana, “Bulk Boundary Condition for Surface Calculations in Density Functional Theory,” *arXiv*:2607.07894 (2026)
2. P. Ray, **S. Bhowmik**, S. R. Kalidindi, P. Suryanarayana, A. J. Medford, “Electronic manifolds for extrapolative alloy discovery,” *accepted Digital Discovery*

3. **S. Bhowmik**, A. Sharma, A. J. Medford, J. E. Pask, P. Suryanarayana, “Ab initio study of strain-driven vacancy clustering in aluminum,” *Phys. Rev. B*, 112, 174105 (2025).
4. **S. Bhowmik**, A. J. Medford, P. Suryanarayana, “Real-space Hubbard-corrected density functional theory,” *J. Chem. Phys.* 163, 234113 (2025).
5. **S. Bhowmik**, J. E. Pask, A. J. Medford, P. Suryanarayana, “Spectral scheme for atomic structure calculations in density functional theory,” *Comput. Phys. Commun.* 308 109448 (2025).
6. **S. Bhowmik**, A. Govind Rajan, “Size and Chemical Environment Control Nanopore Geometry in 2D MoS₂: From Irregular to Triangular Defects,” *Small*, 2504611 (2025).
7. **S. Bhowmik**, J. H. Warner, A. Govind Rajan, “Role of Chemical Etching in the Nucleation of Nanopores in 2D MoS₂: Insights from First-Principles Calculations,” *J. Phys. Chem. C* 127, 14, 6873-6883 (2023).
8. **S. Bhowmik**, A. Govind Rajan, “Chemical Vapor Deposition of 2D Materials: A Review of Modeling, Simulation, and Machine Learning Studies,” *iScience*, 103832 (2022).
9. **S. Bhowmik**, V. Chakraborty, P. Das, “Batch adsorption of indigo carmine on activated carbon prepared from sawdust: A comparative study and optimization of operating conditions using Response Surface Methodology,” *Results in Surfaces and Interfaces* 3, 100011 (2021).

PRESENTATIONS

- **AIChE Annual Meeting (2026, Forthcoming)**: “Beyond Vacuum Padding: Bulk-Boundary Conditions for Accurate Semi-Periodic Density Functional Theory,” Minneapolis, MN, USA. (*Oral*)
- **ACS Spring (2026)**: “Real-space Hubbard-corrected density functional theory and Hubbard parameters from hybrid functionals”, Atlanta, GA, USA. (*Oral*)
- **ACS Spring (2026)**: “From irregular defects to perfect triangles: Tuning nanopore geometry in 2D MoS₂ via size and chemical environment control”, Atlanta, GA, USA. (*Oral*)
- **SES (2025)**: “Ab initio investigation into the aggregation of vacancies and formation of dislocation loops in aluminum,” Atlanta, GA, USA. (*Oral*)
- **GRC (2024)**: “Ab initio investigation into the aggregation of vacancies and formation of dislocation loops in aluminum,” Newry, ME, USA. (*Poster*)
- **GRS (2024)**: “SPARC-atom: An efficient real-space electronic structure calculator for atoms,” Newry, ME, USA. (*Poster*)

HONORS & AWARDS

- **Eckert Graduate Fellowship**, Georgia Tech (2026) · Awarded for research accomplishments.
- **Best Online TA Award**, Georgia Tech (2026) · Exemplary performance as online TA.
- **Shell Outstanding TA Award**, Georgia Tech (2024) · Exemplary performance in core engineering instruction.
- **Best PhD Qualifying Exam Award**, Georgia Tech (2023) · Top student in cohort.
- **Exemplary Academic Achievement Award**, Georgia Tech (2023) · Perfect 4.0/4.0 semester GPA in graduate core coursework.
- **Prof. N. R. Kuloor Memorial Award**, IISc (2022) · Best M.Tech graduate in Chemical Engineering.

TEACHING & LEADERSHIP

Georgia Institute of Technology **Atlanta, GA, USA**
TA · Data Analytics, Quantum Mechanics & Transport Phenomena *2023 – 2025*

- Delivered lectures and designed assessments across three courses; developed [open-access DFT training materials](#); recipient of the **Shell Outstanding TA Award (2024)** and **Best Online TA Award**.

iCoMSE Workshop (10th), Oklahoma State University **Stillwater, OK, USA**
Workshop Instructor: Computational Methodologies Training *June 2025*

- Led hands-on **SPARC** training sessions for high-performance DFT/AIMD simulations.